

Understanding diffusion model from a mathematical view

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What I cannot create, I do not understand.

Outline

- 1 Preliminary: Generating Samples from Probability Distributions
- 2 Matching score for training
- 3 Relations of different models: x_0 , epsilon and score model
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Generate samples from given distribution

Given a distribution $p(x)$, how to sample from the distribution?

- Uniform distribution & Gaussian distribution: easy sampling
- Mixture Gaussian: determine which Gaussian to sample & sample from that Gaussian
- A general sample function?
 - Langevin Markov Chain Monte Carlo sampling (Langevin MCMC)
 - Stein Variational Gradient Descent (SVGD)
 - ...

Score function is all you need?

A typical process of generating new samples from modeling the data distribution

- ① To find the probability function $p(x)$, model $p(x; \theta)$.
- ② Normalization distribution $p(x; \theta) = \frac{1}{C(\theta)} e^{-E(x; \theta)}$ with $C(\theta) \equiv \int_x e^{-E(x; \theta)} dx$.
- ③ Generating samples from above sampling methods.

Recall the Langevin Markov Chain Monte Carlo sampling

$$x_t = x_{t-1} + \frac{\Delta t}{2} \nabla_x \log p(x_{t-1}; \theta) + \sqrt{\Delta t} \epsilon, \quad \epsilon \sim N(0, 1). \quad (1)$$

Here $\nabla_x \log p(x_{t-1}; \theta) = -\nabla_x E(x; \theta)$ is free of the normalization $C(\theta)$.

Define the score function

$$S(x; \theta) \equiv \nabla \log p(x; \theta). \quad (2)$$

More about Langevin MCMC and the score function

Continuous form of the Langevin MCMC

- $\Delta t \rightarrow 0$

$$dx = \frac{1}{2} \nabla_x \log p(x) dt + dB_t, \quad (3)$$

where dB_t is a Brownian motion.

- Given $p(x) = \frac{1}{C} e^{-E}$, we have

$$dx = -\frac{1}{2} \nabla_x E dt + dB_t \quad (4)$$

Langevin MCMC is overdamped Langevin dynamics

- Langevin dynamics (MD)

$$\ddot{x} = -\nabla_x E - \gamma \dot{x} + dB_t, \quad (5)$$

- Given $\gamma \dot{x} \gg \ddot{x}$, we have the overdamped Langevin equation

$$dx = -\frac{1}{\gamma} \nabla_x E + \frac{1}{\gamma} dB_t.$$

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Matching the score by neural networks

Recall the definition

$$S \equiv \nabla_x \log p(x), \quad (7)$$

The key is to matching the score by a neural network

$$J_{ESM} = \frac{1}{2} \mathbb{E}_p \left[\left\| s(x; \theta) - \frac{\partial \log p(x)}{\partial x} \right\|^2 \right], \quad (8)$$

where J_{ESM} is called Explicit Score Matching loss. However, it is not tractable to compute $\frac{\partial \log p(x)}{\partial x}$ from data.

Where should we go?

Implicit Score Matching (ISM) loss

Implicit Score Matching loss (Aapo, 2005) was developed to make a tractable score matching

$$J_{ISM} \equiv \mathbb{E}_p \left[\frac{1}{2} \|s(x; \theta)\|^2 + \nabla \cdot s(x; \theta) \right] = J_{ESM} - C. \quad (9)$$

Proof.

$$\begin{aligned} J_{ESM} &= \frac{1}{2} \mathbb{E}_p \left[\left\| s(x; \theta) - \frac{\partial \log p(x)}{\partial x} \right\|^2 \right] \\ &= \frac{1}{2} \mathbb{E}_p [\|s\|^2] - \mathbb{E}_p \left[s \cdot \frac{\partial \log p}{\partial x} \right] + \frac{1}{2} \mathbb{E}_p \left[\left\| \frac{\partial \log p}{\partial x} \right\|^2 \right] \end{aligned} \quad (10)$$

where

$$\begin{aligned} -\mathbb{E}_p \left[s \cdot \frac{\partial \log p}{\partial x} \right] &= -\int_x p s \cdot \nabla \log p dx = -\int_x p s \cdot \frac{\nabla p}{p} dx \\ &= -\int_x s \cdot \nabla p dx = -\int \nabla \cdot (ps) dx + \int p \nabla \cdot s dx = \mathbb{E}_p [\nabla \cdot s]. \end{aligned} \quad (11)$$

So we have $J_{ISM} = J_{ESM} - C$, where $C \equiv \frac{1}{2} \mathbb{E}_p \left[\left\| \frac{\partial \log p}{\partial x} \right\|^2 \right]$.

Denoising Score Matching (DSM) loss

However, the ISM score is not very stable to optimize, with two reasons

- ① Expectation is performed on the whole distribution;
- ② The loss is negative decreasing to $-C$ with a great quantity, e.g. $-1e5$.

Denoising Score Matching (DSM) loss (Vincent, 2011) is developed to solve this problem

$$J_{DSM} = \frac{1}{2} \mathbb{E}_{p(x, \tilde{x})} [\|s(\tilde{x}; \theta) - \nabla_{\tilde{x}} \log p(\tilde{x}|x)\|^2]. \quad (12)$$

And we can prove that $J_{DSM} = J_{ESM} + C$.

Prove the equivalence $J_{DSM} = J_{ESM} + C$

Proof.

We expand the formula and check the items one by one.

$$\begin{aligned} J_{DSM} &\equiv \frac{1}{2} \mathbb{E}_{p(x, \tilde{x})} [\|s(\tilde{x}; \theta) - \nabla_{\tilde{x}} \log p(\tilde{x}|x)\|^2] \\ &= \frac{1}{2} \mathbb{E}_{p(\tilde{x})} [\|s(\tilde{x})\|^2] - \mathbb{E}_{p(x, \tilde{x})}[s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x}|x)] + \frac{1}{2} \mathbb{E}_{p(x, \tilde{x})} [\|\nabla_{\tilde{x}} \log p(\tilde{x}|x)\|^2] \end{aligned} \quad (13)$$

And we can prove that the cross term of J_{ESM} and J_{DSM} is equal.

$$\begin{aligned} \mathbb{E}_p(\tilde{x})[s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x})] &= \int_{\tilde{x}} p(\tilde{x}) s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x}) d\tilde{x} \\ &= \int_{\tilde{x}} s(\tilde{x}) \cdot \nabla_{\tilde{x}} p(\tilde{x}) d\tilde{x} = \int_{\tilde{x}} \int_x p(x) s(\tilde{x}) \cdot \nabla_{\tilde{x}} p(\tilde{x}|x) d\tilde{x} dx \\ &= \int_{\tilde{x}} \int_x p(\tilde{x}|x) p(x) s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x}|x) d\tilde{x} dx = \mathbb{E}_{p(\tilde{x}, x)} [s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x}|x)]. \end{aligned} \quad (14)$$

Finally we have

$$J_{DSM} = J_{ESM} + C \quad \frac{1}{2} \mathbb{E}_{p(\tilde{x})} [\|s(\tilde{x})\|^2] - \mathbb{E}_{p(x, \tilde{x})}[s(\tilde{x}) \cdot \nabla_{\tilde{x}} \log p(\tilde{x}|x)] + \frac{1}{2} \mathbb{E}_{p(x, \tilde{x})} [\|\nabla_{\tilde{x}} \log p(\tilde{x}|x)\|^2] = \frac{1}{2} \mathbb{E}_{p(x, \tilde{x})} [\|s(\tilde{x}) - \nabla_{\tilde{x}} \log p(\tilde{x}|x)\|^2] + C \quad (15)$$

DSM is all you need?

Problem arises when $\tilde{x} \rightarrow x$. We can prove that when Gaussian noise added by $p(\tilde{x}|x) = N(\alpha x, \sigma^2)$,

$$J_{DSM} \rightarrow \infty, \text{ if } \tilde{x} \rightarrow x. \quad (16)$$

Proof.

Given the Gaussian noise added, we have

$$\nabla_{\tilde{x}} \log p(\tilde{x}|x) = \nabla_{\tilde{x}} \log e^{-\frac{(\tilde{x}-\alpha x)^2}{2\sigma^2}} = -\frac{\tilde{x} - \alpha x}{\sigma^2}. \quad (17)$$

And we can show that the following formula goes to infinity

$$\begin{aligned} \mathbb{E}_{p(x, \tilde{x})} [\|\nabla_{\tilde{x}} \log p(\tilde{x}|x)\|^2] &= \mathbb{E}_{p(x, \tilde{x})} [\|\frac{\tilde{x} - \alpha x}{\sigma^2}\|^2] \\ &= \mathbb{E}_{p(x)} \mathbb{E}_{p(\tilde{x}|x)} [\|\frac{\tilde{x} - \alpha x}{\sigma^2}\|^2] = \mathbb{E}_{\epsilon \sim N(0,1)} [\|\frac{\epsilon}{\sigma}\|^2] \rightarrow \infty \text{ if } \sigma \rightarrow 0. \end{aligned} \quad (18)$$

Finally, we know that

$$J_{DSM} = J_{ESM} + C \rightarrow \infty \text{ as } C \rightarrow \infty \text{ when } \tilde{x} \rightarrow x. \quad (19)$$

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Revisit the adding noise from x to $\tilde{x}$

Gaussian transition kernel for adding noise from x to $\tilde{x}$

$$\tilde{x} = \alpha x + \sigma \epsilon, \quad (20)$$

which equivalent to the Gaussian transition kernel

$$p(\tilde{x}|x) = N(\tilde{x}; \alpha x, \sigma^2), \quad (21)$$

where $\epsilon \sim N(0, 1)$ and

$$N(\tilde{x}; \alpha x, \sigma^2) = C e^{-\frac{\|\tilde{x} - \alpha x\|^2}{2\sigma^2}}. \quad (22)$$

And

$$\epsilon = \frac{\tilde{x} - \alpha x}{\sigma}. \quad (23)$$

Questions arise: how can we get x given $\tilde{x}$?

Tweedie's Formula

Tweedie's Formula (Bayes statistics) states (Robbins, 1956)

Theorem

Given random variables x, y , $y \sim N(x, \sigma^2 I)$, i.e., $p(y|x) = N(x, \sigma^2)$, the expectation of x could be given by

$$\mathbb{E}[x|y] = y + \sigma^2 \nabla \log p(y). \quad (24)$$

Let

$$x \leftarrow \alpha x, \quad y \leftarrow \tilde{x}, \quad (25)$$

we have

$$s \equiv \nabla_{\tilde{x}} \log p(\tilde{x}) = -\frac{\tilde{x} - \alpha \mathbb{E}[x|\tilde{x}]}{\sigma^2} = -\frac{1}{\sigma} \mathbb{E}\left[\frac{\tilde{x} - \alpha x}{\sigma} | \tilde{x}\right] = -\frac{\mathbb{E}[\epsilon|\tilde{x}]}{\sigma}, \quad (26)$$

where we have used the fact $\epsilon = \frac{\tilde{x} - \alpha x}{\sigma}$.

Another insight: from the definition of the score S

Recall that adding noise by $\tilde{x} = \alpha x + \sigma \epsilon$,

$$p(\tilde{x}|x) = Ce^{-\frac{\|\tilde{x}-\alpha x\|^2}{2\sigma^2}}. \quad (27)$$

Hence

$$s(\tilde{x}) = \nabla_{\tilde{x}} \log p(\tilde{x}) = \frac{\nabla_{\tilde{x}} p(\tilde{x})}{p(\tilde{x})} = \frac{\int_x \nabla_{\tilde{x}} p(\tilde{x}|x) p(x) dx}{p(\tilde{x})}, \quad (28)$$

From Eq. (27), we have

$$s(\tilde{x}) = \frac{\int_x \nabla_{\tilde{x}} \log p(\tilde{x}|x) p(\tilde{x}|x) p(x) dx}{p(\tilde{x})} = \int_x \nabla_{\tilde{x}} \log p(\tilde{x}|x) p(x|\tilde{x}) dx, \quad (29)$$

leading to

$$s(\tilde{x}) = \mathbb{E}_x[\nabla_{\tilde{x}} \log p(\tilde{x}|x)|\tilde{x}] = \mathbb{E}_x[-\frac{\tilde{x} - \alpha x}{\sigma^2}|\tilde{x}] = -\frac{\tilde{x} - \alpha \mathbb{E}[x|\tilde{x}]}{\sigma^2}.$$

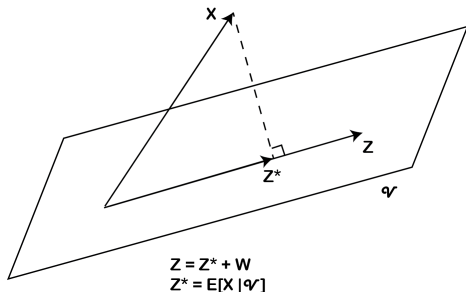
Theoretical Support: Conditional Expectation

Theorem

Let X be an integrable random variable. Then for each σ -algebra $\mathcal{V}$ and $Z \in \mathcal{V}$, $Z^* = \mathbb{E}(X|\mathcal{V})$ solves the least square problem

$$\operatorname{argmin}_{Z \in \mathcal{V}} \|Z - X\|,$$

where $\|Z\| = \left(\int Z^2 dP\right)^{\frac{1}{2}}$.



Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $L^2 := L^2(\Omega, \mathcal{F}, \mathbb{P})$ with $\langle U, V \rangle = \mathbb{E}[UV]$. Fix a sub- σ -field $\mathcal{G} \subset \mathcal{F}$ and denote the closed subspace

$$\mathcal{V} := L^2(\mathcal{G}) = \{Z \in L^2 : Z \text{ is } \mathcal{G}\text{-measurable}\}.$$

Claim. For $X \in L^2$,

$$Z^* := \mathbb{E}[X \mid \mathcal{G}] \in \mathcal{V} \quad \text{and} \quad Z^* = \arg \min_{Z \in \mathcal{V}} \mathbb{E}[(Z - X)^2].$$

Proof of the claim

For any $Z \in \mathcal{V}$,

$$\langle X - Z^*, Z \rangle = \mathbb{E}[(X - Z^*)Z] = \mathbb{E}(\mathbb{E}[(X - Z^*)Z \mid \mathcal{G}]) = \mathbb{E}(Z \mathbb{E}[X - Z^* \mid \mathcal{G}]) = 0.$$

Hence $X - Z^* \perp \mathcal{V}$.

Let any $Z \in \mathcal{V}$ be written as $Z = Z^* + W$ with $W \in \mathcal{V}$. Then

$$\|X - Z\|_2^2 = \|X - (Z^* + W)\|_2^2 = \|X - Z^*\|_2^2 + \|W\|_2^2 \geq \|X - Z^*\|_2^2,$$

because $\langle X - Z^*, W \rangle = 0$. Equality $\Leftrightarrow W = 0 \Leftrightarrow Z = Z^*$, proving uniqueness and the minimizer claim.

Revisit several models

① Score model

$$Loss_{score} = \|s_{\theta}(\tilde{x}) - \frac{\epsilon}{\sigma}\|^2; \quad (31)$$

② X prediction model (denoising model, x_0 model)

$$Loss_{x_0} = \|x_{\theta}(\tilde{x}) - x\|^2, \quad (32)$$

with

$$s(\tilde{x}) = -\frac{\tilde{x} - \alpha x_{\theta^*}(\tilde{x})}{\sigma^2} \quad (33)$$

where $x_{\theta}(\tilde{x}) \rightarrow x_{\theta^*}(\tilde{x}) \equiv \mathbb{E}[x|\tilde{x}]$;

③ ϵ prediction model

$$Loss_{\epsilon} = \|\epsilon_{\theta}(\tilde{x}) - \epsilon\|^2, \quad (34)$$

with

$$s(\tilde{x}) = -\frac{\epsilon_{\theta^*}(\tilde{x})}{\sigma}, \quad (35)$$

where $\epsilon_{\theta}(\tilde{x}) \rightarrow \epsilon_{\theta^*}(\tilde{x}) \equiv \mathbb{E}[\epsilon|\tilde{x}]$.

The consistency of the three modeling

As stated before, we have the following connection in the optimal (θ^*) case

$$s_{\theta^*}(\tilde{x}) = -\frac{\tilde{x} - \alpha x_{\theta^*}(\tilde{x})}{\sigma^2} = -\frac{\epsilon_{\theta^*}(\tilde{x})}{\sigma}, \quad (36)$$

by the above conditional expectation.

Hence, in the real modeling of designing the loss, we use the connection of three models without reaching the optimal parameter

$$s_{\theta}(\tilde{x}) = -\frac{\tilde{x} - \alpha x_{\theta}(\tilde{x})}{\sigma^2} = -\frac{\epsilon_{\theta}(\tilde{x})}{\sigma}, \quad (37)$$

and substitute each other form to the loss definition above, we can recover all of the losses given by different models.

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Key Identity: Jacobian Switch (chain rule between $\mathbf{x}_0$ and $\mathbf{x}_t$)

The Gaussian transition density is

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \alpha_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}) \propto \exp\left(-\frac{\|\mathbf{x}_t - \alpha_t \mathbf{x}_0\|^2}{2\sigma_t^2}\right).$$

A basic but crucial identity:

$$\nabla_{\mathbf{x}_t} q(\mathbf{x}_t | \mathbf{x}_0) = -\frac{1}{\alpha_t} \nabla_{\mathbf{x}_0} q(\mathbf{x}_t | \mathbf{x}_0), \quad (\alpha_t > 0).$$

Proof sketch. Differentiating the exponent w.r.t. $\mathbf{x}_t$ and w.r.t. $\mathbf{x}_0$ shows

$\nabla_{\mathbf{x}_t} \|\mathbf{x}_t - \alpha_t \mathbf{x}_0\|^2 = 2(\mathbf{x}_t - \alpha_t \mathbf{x}_0)$, $\nabla_{\mathbf{x}_0} \|\mathbf{x}_t - \alpha_t \mathbf{x}_0\|^2 = -2\alpha_t(\mathbf{x}_t - \alpha_t \mathbf{x}_0)$, thus the factor $-1/\alpha_t$.

Theorem 1: Score as Conditional Force Expectation

Statement. With $p(\mathbf{x}_0) \propto e^{-E(\mathbf{x}_0)/k_B T}$ and $\mathbf{x}_t = \alpha_t \mathbf{x}_0 + \sigma_t \epsilon$,

$$\boxed{\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) = \frac{1}{\alpha_t} \mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t} \left[\frac{\mathbf{F}(\mathbf{x}_0)}{k_B T} \right]}, \quad \mathbf{F}(\mathbf{x}_0) = -\nabla_{\mathbf{x}_0} E(\mathbf{x}_0).$$

Derivation.

$$\begin{aligned} \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) &= \frac{\nabla_{\mathbf{x}_t} \int p(\mathbf{x}_0) q(\mathbf{x}_t | \mathbf{x}_0) d\mathbf{x}_0}{p(\mathbf{x}_t)} \\ &= \frac{1}{p(\mathbf{x}_t)} \int p(\mathbf{x}_0) \nabla_{\mathbf{x}_t} q(\mathbf{x}_t | \mathbf{x}_0) d\mathbf{x}_0 \\ &\stackrel{(*)}{=} \frac{1}{p(\mathbf{x}_t)} \int p(\mathbf{x}_0) \left(-\frac{1}{\alpha_t} \right) \nabla_{\mathbf{x}_0} q(\mathbf{x}_t | \mathbf{x}_0) d\mathbf{x}_0 \\ &= -\frac{1}{\alpha_t} \frac{1}{p(\mathbf{x}_t)} \int q(\mathbf{x}_t | \mathbf{x}_0) \nabla_{\mathbf{x}_0} p(\mathbf{x}_0) d\mathbf{x}_0 \\ &= -\frac{1}{\alpha_t} \int \underbrace{\frac{p(\mathbf{x}_0) q(\mathbf{x}_t | \mathbf{x}_0)}{p(\mathbf{x}_t)}}_{p(\mathbf{x}_0|\mathbf{x}_t)} \nabla_{\mathbf{x}_0} \log p(\mathbf{x}_0) d\mathbf{x}_0 \\ &= \frac{1}{\alpha_t} \mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t} \left[\frac{-\nabla_{\mathbf{x}_0} E(\mathbf{x}_0)}{k_B T} \right] = \frac{1}{\alpha_t} \mathbb{E}_{\mathbf{x}_0|\mathbf{x}_t} \left[\frac{\mathbf{F}(\mathbf{x}_0)}{k_B T} \right]. \end{aligned}$$

Experiments on first 1, 000 frames for LJ data

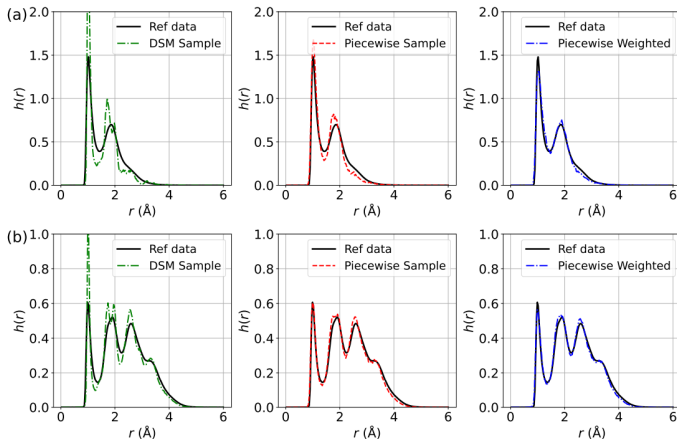


Figure 1: The distribution of interatomic distances $h(r)$ for LJ potential using biased training data with a sample size of 500. (a) Comparison of $h(r)$ for DSM, Piecewise, and Piecewise Weighted losses (from left to right) against reference data for LJ-13; (b) Comparison of $h(r)$ for DSM, Piecewise, and Piecewise Weighted losses for LJ-55.

Experiments on first 5, 000 frames for MD reference trajectories

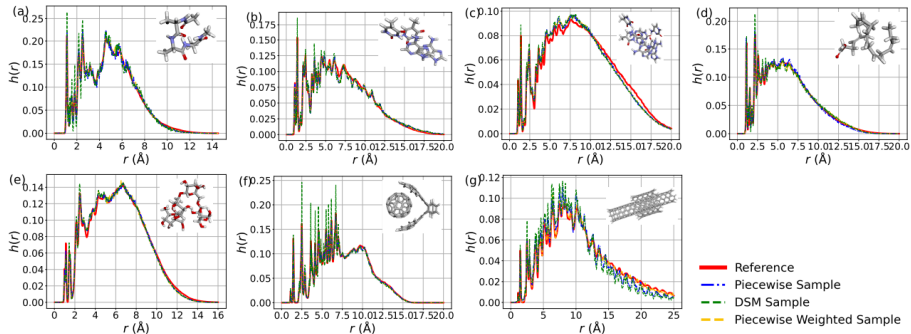


Figure 3: The distribution of interatomic distances (r (Å)) for (a) Ac-Ala3-NHMe, (b) DNA base pair (AT-AT), (c) DNA base pair (AT-AT-CG-CG), (d) Docosahexaenoic acid, (e) Stachyose, (f) Buckyball catcher, (g) Dw nanotube in MD22 dataset. The insets display the ball-and-stick representations of these molecules.

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Score Matching Langevin Dynamic (SMLD) (Song, 2019) adding noise in a following manner

$$p(\tilde{x}_i|x) = N(\tilde{x}_i; x, \sigma_i^2) \equiv Ce^{-\frac{\|\tilde{x}_i - x\|^2}{2\sigma_i^2}}, \quad (38)$$

where a geometric sequence $\sigma_1 > \sigma_2 > \dots > \sigma_T \approx 0$ is given to add noise with different level. In a random variable view,

$$\tilde{x}_i = x + \sigma_i \epsilon \quad (39)$$

for different σ_i to get different noised random variable $\tilde{x}_i$.

Training object J_{DSM} is given by

$$J_{DSM} = \frac{1}{2} \mathbb{E}_{\sigma_i} \mathbb{E}_{p(x)} \mathbb{E}_{p(\tilde{x}_i|x)} \left[\left\| s_{\theta}(\tilde{x}_i, \sigma_i) + \frac{\tilde{x}_i - x}{\sigma_i^2} \right\|^2 \right]. \quad (40)$$

Anneled Langevin MCMC for sampling

$$x_t = x_{t-1} + \frac{\Delta t}{2} S_{\theta}(x_{t-1}) + \sqrt{\Delta t} \epsilon, \quad \epsilon \sim N(0, 1).$$

denoising diffusion probabilistic models (DDPM) (Ho, 2020)

- ① Derived from the Evidence Lower BOund (ELBO), **Not** score matching.
- ② First provide adding noise and denoise in the forward and reverse process.

The main procedure

- ① Adding noise

$$x_t = \sqrt{1 - \beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon; \quad (42)$$

- ② Transition kernel

$$p(x_t|x_0) = N(x_t; \alpha_t x_0, \sigma_t^2), \quad (43)$$

where $\alpha_t = \prod_{i=1}^t \sqrt{1 - \beta_i}$ and $\sigma_t^2 = 1 - \alpha_t^2$.

The main procedure (Continued)

- 1 Adding noise

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \epsilon; \quad (44)$$

- 2 Transition kernel

$$p(x_t | x_0) = N(x_t; \alpha_t x_0, \sigma_t^2), \text{ i.e., } x_t = \alpha_t x_0 + \sigma_t \epsilon, \quad (45)$$

where $\alpha_t = \prod_{i=1}^t \sqrt{1 - \beta_i}$ and $\sigma_t^2 = 1 - \alpha_t^2$.

- 3 Training Loss

$$\mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{p(x)} \mathbb{E}_{p(x_t|x)} [\|\epsilon_\theta(x_t, t) - \epsilon\|^2], \quad (46)$$

where we have used the approximation $x_0 = \frac{x_t - \sigma_t \epsilon}{\alpha_t}$.

- 4 Sampling

$$x_{t-1} = \frac{1}{\sqrt{1 - \beta_t}} (x_t + \beta_t s_\theta(x_t, t)) + \sqrt{\beta_t} \epsilon_t, \quad t = T, T-1, \dots, 1. \quad (47)$$

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The continuous version of SMLD

Recall that

$$x_t = x_0 + \sigma_t \epsilon, \quad (48)$$

we can prove that

$$x_t = x_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \epsilon. \quad (49)$$

Proof.

We can prove the formula by a Recurrence

$$\begin{aligned} x_t &= x_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \epsilon \\ &= x_{t-2} + \sqrt{\sigma_{t-1}^2 - \sigma_{t-2}^2} \epsilon + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \epsilon \\ &= x_{t-2} + \sqrt{\sigma_t^2 - \sigma_{t-2}^2} \epsilon \\ &= \dots = x_0 + \sigma_t \epsilon, \quad \text{where } \sigma_0 \approx 0. \end{aligned} \quad (50)$$

The continuous version of SMLD

From

$$x_t = x_{t-1} + \sqrt{\sigma_t^2 - \sigma_{t-1}^2} \epsilon, \quad (51)$$

we have

$$\begin{aligned} \Delta x_t &= \sqrt{\frac{\sigma_t^2 - \sigma_{t-1}^2}{\Delta t}} \sqrt{\Delta t} \epsilon \\ &= \sqrt{\frac{\sigma_t^2 - \sigma_{t-1}^2}{\Delta t}} \Delta B_t. \end{aligned} \quad (52)$$

Further, let $\Delta t \rightarrow 0$, we have

$$dx = \sqrt{\frac{d\sigma_t^2}{dt}} dB_t. \quad (53)$$

The continuous version of DDPM

Recall that

$$x_t = \sqrt{1 - \beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon. \quad (54)$$

Similarly

$$x_t \approx (1 - \frac{1}{2}\beta_t)x_{t-1} + \sqrt{\beta_t}\epsilon, \quad (55)$$

hence, we have

$$\begin{aligned} \Delta x_t &= -\frac{1}{2}\tilde{\beta}_t x_{t-1} \Delta t + \sqrt{\tilde{\beta}_t} \sqrt{\Delta t} \epsilon, \quad \text{where } \tilde{\beta}_t = \frac{\beta_t}{\Delta t} \\ &= -\frac{1}{2}\tilde{\beta}_t x_{t-1} \Delta t + \sqrt{\tilde{\beta}_t} \Delta B_t. \end{aligned} \quad (56)$$

Further, let $\Delta t \rightarrow 0$, we finally obtain

$$dx = -\frac{1}{2}\tilde{\beta}_t x dt + \sqrt{\tilde{\beta}_t} dB_t. \quad (57)$$

Summarizing the form of continuous DDPM and SMLD, we summarize the adding noise process as

$$dx = f(x, t)dt + g(t)dB_t, \quad (58)$$

where the corresponding distribution $p(x, t)$ satisfying the following Fokker-Planck equation

$$\frac{\partial p(x, t)}{\partial t} + \nabla \cdot (f(x, t)p(x, t)) - \frac{1}{2}g^2(t)\Delta p(x, t) = 0. \quad (59)$$

Training DSM object

$$J_{DSM} = \mathbb{E}_{t \sim U[0,1]} \mathbb{E}_{p(x_0)} \mathbb{E}_{p(x_t|x_0)} [\|s_\theta(x_t, t) - \nabla_{x_t} \log p(x_t|x_0)\|^2]. \quad (60)$$

VE & VP adding noise

To training the Denoising Score Match loss J_{DSM} above, the key is to add noise by the transition kernel

$$p(\tilde{x}|x) \equiv p(x(t)|x(0)) = N(x(t); \tilde{\alpha}_t x(0), \tilde{\sigma}_t^2), \quad (61)$$

from which we can obtain the noised data

$$x(t) = \tilde{\alpha}_t x(0) + \tilde{\sigma}_t \epsilon, \quad \epsilon \sim N(0, 1). \quad (62)$$

The question becomes: Given VE & VP

$$dx = \sqrt{\frac{d[\sigma^2]}{dt}} dB_t, \quad dx = -\frac{1}{2} \tilde{\beta}_t x dt + \sqrt{\tilde{\beta}_t} dB_t. \quad (63)$$

how to derive $p(x(t)|x(0))$, i.e., $p(\tilde{x}|x)$ for adding noise?

VE & VP Transition kernel

The summarizing form of VE & VP is

$$dx = h(t)xdt + g(t)dB_t, \quad (64)$$

where $h(t) = 0$ for VE and $h(t) = -\frac{1}{2}\tilde{\beta}(t)$ for VP.

Lemma

Let $\mu(t) = \mathbb{E}[x(t)]$ and $\Sigma(t) = \mathbb{E}[(x - \mu(t))(x - \mu(t))^T]$, we have the following formula

$$\frac{d\mu(t)}{dt} = h(t)\mu(t), \quad \frac{d\Sigma(t)}{dt} = 2h(t)\Sigma(t) + g^2(t)I. \quad (65)$$

The proof of the Lemma can be found in a stochastic differential equation textbook such as Applied Stochastic Differential Equations, Särkkä and Solin, 2019.

VE & VP Transition kernel

Lemma

For VE, the transition kernel is given by the following formula

$$P(x(t)|x(0)) = N(x(t); x(0), \sigma^2(t) - \sigma^2(0)). \quad (66)$$

For VE,

$$\tilde{\alpha}_t \equiv 1, \quad \tilde{\sigma}_t^2 \equiv \sigma^2(t) - \sigma^2(0) \approx \sigma^2(t). \quad (67)$$

Lemma

For VP, the transition kernel is given the following formula

$$p(x(t)|x(0)) = N\left(x(t); x(0)e^{-\frac{1}{2} \int_0^t \tilde{\beta}(s) ds}, 1 - e^{-\int_0^t \tilde{\beta}(s) ds}\right). \quad (68)$$

For VP,

$$\tilde{\alpha}_t \equiv e^{-\frac{1}{2} \int_0^t \tilde{\beta}(s) ds}, \quad \tilde{\sigma}_t^2 \equiv 1 - e^{-\int_0^t \tilde{\beta}(s) ds}. \quad (69)$$

Proof of VE transition kernel

Proof.

For VE, by a simple calculation we have

$$\frac{d\mu(t)}{dt} = 0, \quad \frac{d\Sigma(t)}{dt} = \frac{d[\sigma^2(t)]}{dt}, \quad (70)$$

Hence, it is easy to obtain the following form

$$\mu(t) = \mu(0) = x(0), \quad \Sigma(t) = \sigma^2(t) - \sigma^2(0). \quad (71)$$

Here we have used the fact

$$\mu(0) = x(0), \quad \Sigma(0) = 0 \quad (72)$$

as $x(0)$ is taken as a constant, not a random variable. And we obtain

$$P(x(t)|x(0)) = N(x(t); x(0), \sigma^2(t) - \sigma^2(0)). \quad (73)$$

Proof of VP transition kernel

Proof.

For VP, by a simple calculation we have

$$\frac{d\mu(t)}{dt} = -\frac{1}{2}\beta(t)\mu(t), \quad \frac{d\Sigma(t)}{dt} = -\beta(t)\Sigma(t) + \beta(t). \quad (74)$$

Multiply the exponential term on the both sides, we have

$$\begin{aligned} \frac{d\mu(t)}{dt} e^{\int_0^t \frac{1}{2}\beta(s)ds} + \frac{1}{2}\beta(t)\mu(t)e^{\int_0^t \frac{1}{2}\beta(s)ds} &= 0, \\ \frac{d\Sigma(t)}{dt} e^{\int_0^t \beta(s)ds} + \beta(t)\Sigma(t)e^{\int_0^t \beta(s)ds} &= \beta(t)e^{\int_0^t \beta(s)ds}. \end{aligned} \quad (75)$$

Hence, we have the following form

$$\frac{d}{dt}(\mu(t)e^{\int_0^t \frac{1}{2}\beta(s)ds}) = 0, \quad \frac{d}{dt}(\Sigma(t)e^{\int_0^t \beta(s)ds}) = \frac{d}{dt}e^{\int_0^t \beta(s)ds}. \quad (76)$$

By a direct calculation, we can obtain

$$p(x(t)|x(0)) = N\left(x(t); x(0)e^{-\frac{1}{2}\int_0^t \beta(s)ds}, 1 - e^{-\int_0^t \beta(s)ds}\right). \quad (77)$$

Revisit adding noise

The **equivalence** of the adding noise

- ① From the stochastic process, denote the variable x_t , we have

$$dx_t = f(x, t)dt + g(t)dB_t, \quad (78)$$

with corresponding discrete form (Euler-Maruyama scheme)

$$x_{t+\Delta t} = x_t + f(x_t, t)\Delta t + g(t)\sqrt{\Delta t}\epsilon, \quad \epsilon \sim N(0, 1). \quad (79)$$

- ② From transition kernel perspective,

$$p(x_t) = \int_{x_0} p(x_0)p(x_t|x_0)dx_0, \quad (80)$$

where

$$p(x_t|x_0) = N(x_t; \mu_t, \Sigma_t) = N(x_t; \tilde{\alpha}_t x_0, \tilde{\sigma}_t^2) = \frac{1}{C} e^{-\frac{\|x_t - \mu_t\|^2}{2\Sigma_t}}.$$

Training recipe

The training for VE follows the following process (VP is in a similar way)

- ① Random choose one data $x \sim p_{data}$;
- ② Random sample $t \sim U[0, 1]$;
- ③ Random sample a white noise $\epsilon \sim N(0, 1)$;
- ④ Adding noise

$$x_t = \tilde{\alpha}_t x + \tilde{\sigma}_t \epsilon, \quad (82)$$

where the mean value $\tilde{\alpha}_t x = x(0)$ and standard deviation $\tilde{\sigma}_t = \sqrt{\sigma^2(t) - \sigma^2(0)} \approx \sigma(t)$ in VE referring to (66). VP is similar with the mean and standard deviation from its transition kernel (68);

- ⑤ Compute the Loss by summation the following norm over x , t and ϵ

$$\|s_\theta(x_t, t) - \epsilon / \tilde{\sigma}_t\|.$$

Inference: Reverse denoising process

The reverse denoising process is given by

$$dx = (f(x, t) - g^2 \nabla_x \log p(x, t))dt + g(t)dB_t, \quad (84)$$

or

$$dx = (f(x, t) - \frac{1}{2}g^2 \nabla_x \log p(x, t))dt, \quad (85)$$

or

$$dx = (f(x, t) - \frac{3}{2}g^2 \nabla_x \log p(x, t))dt + \sqrt{2}g(t)dB_t, \quad (86)$$

$$\dots \quad (87)$$

Due to obeying the same Fokker-Planck equation for $p(x, t)$

$$\frac{\partial p(x, t)}{\partial t} + \nabla \cdot (f(x, t)p(x, t)) - \frac{1}{2}g^2(t)\Delta p(x, t) = 0. \quad (88)$$

Inference: Reverse denoising process (proof)

In this slide, we only prove the first reverse sampling form (84) and other cases can be done in a similar approach.

Proof.

The Fokker-Planck equation of the forward process (58) is given by (59) as

$$\frac{\partial p}{\partial t} + \nabla \cdot (f(x, t)p) - \frac{1}{2}g^2 \Delta p = 0. \quad (89)$$

Let $t = T - \tau$, we have

$$\frac{\partial p}{\partial \tau} - \nabla \cdot (f(x, T - \tau)p) + g^2 \Delta p - \frac{1}{2}g^2 \Delta p = 0. \quad (90)$$

By $\nabla \cdot \nabla = \Delta$ and $\nabla \log p = \nabla p / p$, we have

$$\frac{\partial p}{\partial \tau} - \nabla \cdot \left((f(x, T - \tau) - g^2 \nabla \log p) p \right) - \frac{1}{2}g^2 \Delta p = 0. \quad (91)$$

Hence, we obtain the corresponding SDE form $dx = -(f(x, T - \tau) - g^2 \nabla \log p)d\tau + g dB_\tau$. By a substitution $\tau = T - t$, we finally get $dx = (f(x, t) - g^2 \nabla \log p)dt + g dB_t$. □

Inference recipe

The inference for VE follows the following process (VP is in a similar way)

- 1 Random generate a noise at time $t = 1$ as $x(1) \sim N(0, \sigma_{max}^2)$;
- 2 Integrate the reverse sampling equation

$$dx = (f(x, t) - g^2(t)s_\theta(x, t)) dt + g(t)dB_t \quad (92)$$

or

$$dx = (f - \frac{1}{2}g^2s_\theta)dt \quad (93)$$

over the time span $[0, 1]$ to get $x(0)$.

- 3 Corrector process: at each internal value $x(t)$, we can relax the value $x(t)$ by a corrector, which takes the Langevin MCMC process.

The Corrector: revisit Langevin MCMC

The Langevin diffusion process is given by

$$x_i = x_{i-1} + \frac{\Delta t}{2} \nabla_x \log p(x_{i-1}) + \sqrt{\Delta t} \epsilon, \quad \epsilon \sim N(0, 1). \quad (94)$$

We can prove $\pi(x)$ of the obtained data points $\{x_i\}_{i=1}^N$ converges to $p(x)$.

Proof.

The continuous form of above scheme is given by

$$dx = \frac{1}{2} \nabla \log p(x) dt + dB_t. \quad (95)$$

The corresponding Fokker-Planck equation is written as

$$\frac{\partial \pi(x, t)}{\partial t} + \nabla \cdot \left(\frac{1}{2} \nabla \log p(x) \pi(x, t) \right) - \frac{1}{2} \Delta \pi(x, t) = 0. \quad (96)$$

With a enough long time evolution, the distribution converges to a stable distribution where

$$\frac{\partial \pi(x, t)}{\partial t} = 0, \quad t \rightarrow \infty. \quad (97)$$

The Corrector: revisit Langevin MCMC

Proof (Cont).

Hence we have

$$\nabla \cdot (\pi \nabla \log p - \nabla \pi) = 0, \quad \forall x. \quad (98)$$

Therefore, we have

$$\nabla \log p = \nabla \log \pi, \quad (99)$$

given that $\pi > 0$ almost true. Hence we have

$$\pi^*(x) = p(x). \quad (100)$$

where π^* is the stable distribution of $\pi(x, t)$.

Outline

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- 2 Matching score for training
- 3 Relations of different models: x_0 , epsilon and score model
- 4 Potential Score Matching loss
- 5 Diffusion model: from theories to implementation
- 6 Continuous version of Diffusion Model
- 7 Known and unknown questions**
- 8 More papers
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 - DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Models
 - Consistency Models

Questions

If you are interested in the questions, please feel free to write an email to me (hpp@bimsa.cn) with your comments.

1. Please prove that the forms (84)-(86) are equivalent with same marginal distribution $p(x, t)$ given the same initial condition;
2. Please prove that the score function $s(x, t)$ satisfying the following formula

$$\frac{\partial s}{\partial t} = \nabla(-\nabla \cdot f - \langle f, s \rangle + \frac{1}{2}g^2\|s\|^2 + \frac{1}{2}g^2\langle \nabla, s \rangle); \quad (101)$$

3. Why ISM loss is not commonly used like DSM loss in the diffusion model nowadays? Please make your comments.

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Welcome inputs

Any comments?

Welcome your inputs!